FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, PACKING, VEGETABLE TANNED,

CHROME TANNED, CHROME-VEGETABLE RETANNED

This amendment, which forms a part of Federal Specification KK-L-164A, dated December 8, 1969, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

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3.4.6 Delete and substitute:

"3.4.6 Shrinkage temperature (tannage I). When tested as specified in 4.3, tannage I leather shall have a minimum shrinkage temperature of 75°C.

3.4.6.1 Shrinkage temperature (tannages II and III). When tested as specified in 4.3 tannages II and III leathers shall not shrink when subjected to a temperature of 98°C ± 0.5°C for a period of 30 seconds."

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Table V, in characteristic column, after "Shrinkage temperature". Add "(tannage I)".

Table V Add the following in columns 1 through 5 respectively:

"Shrinkage temperature (tannages II and III) 3.4.6.1 4.3.3 X -"

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Add the following new paragraph:

"4.3.3 Shrinkage temperature. Method 7011 of FED-STD-311 shall be used with the following exceptions:

a. Prior to mounting the test specimen, no more than fifteen specimens shall be placed in a 1000 milliliter suction flask containing approximately 300 milliliters of water at room temperature. A vacuum shall be applied for

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15 to 60 seconds, then released for 15 to 60 seconds. The intermittent vacuum shall be applied until all the specimens settle to the bottom of the flask. The flask shall be gently shaken each time the vacuum is released to prevent specimens that are ready to sink from being entangled with other specimens. One hour after the last release of vacuum, the first specimen shall be removed from the flask and mounted in the shrinkmeter. The remaining specimens shall remain in the water from the suction flask until they are tested. The time lapse between the time when the first specimen is removed from the suction flask and testing of the last specimen is completed shall not exceed six hours.

b. Each specimen shall be fastened in the two clamps of the apparatus and completely immersed in the heating medium. The indicator shall be adjusted to a reference point by taking the slack out of the specimen. (NOTE: If the specimen should lengthen before it contracts as the temperature of the heating medium is being increased at the rate of 3° to 5°C per minute, the reference point at which the indicator comes to rest shall be the new reference point for observing shrinkage.)

c. The thermometer shall be nitrogen filled over mercury, have a white background and a scale from -10°C to 101°C or -1°C to 101°C, graduated in 1°C or finer with a $\pm$ 0.5°C tolerance.

d. After the specimen has been fastened in the two clamps, the specimen need not be placed in the heating medium for 10 minutes before applying heat.

e. A variable transformer capable of controlling the rate of heating (3°C to 5°C per minute) and capable of maintaining a temperature of 98°C $\pm$ 0.5°C shall be used.

f. The temperature of the heating medium shall be raised to 98°C and the temperature of the heating medium shall be kept at 98 $\pm$ 0.5°C for 30 seconds by adjusting the transformer. Any specimen that shrinks prior to reaching 98°C shall be reported as a failure along with the temperature at which the specimen failed. Any specimen that shrinks during the 30 second period shall be reported as a test failure. The number of seconds that elapse between reaching the temperature of 98°C and shrinkage of the specimen shall be reported."

Custodians:

Army - GL
Navy - SA
Air Force - 82

User activities:

Navy - AS, OS, MC

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

AGK-ARS
DC-DCG
GSA-FSS

Project No. 8330-0123

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